

Supplementary material for PLECTA: Overlap-aware instance segmentation of dense strand networks from binary masks

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S1 Instance segmentation: cost function, gates and schedule

This section gives the arithmetic behind Section 2.4 of the article. Values are those of the released configuration, selected exclusively on development data.

S1.1 Skeleton graph

The mask M is binarised at > 0 , skeletonised, and spurs no longer than 3px are removed iteratively. Skeleton pixels of degree at least three form junction clusters, and adjacent clusters merge into nodes: connectors up to 5px are absorbed into one node, connectors up to 14px merge nearby nodes while remaining real arms, and free debris up to 2px is absorbed. Each remaining connected path is stored as an ordered arm carrying one stub at each end.

S1.2 Stub frames

A stub frame holds its tip, outward tangent and signed curvature, estimated by a least-squares arclength polynomial fit read outward from the tip. The fit is quadratic when the span reaches 18px and linear otherwise, and a frame counts as reliable only with at least three samples spanning at least 3px. Local frames use 24px of support; once links form chains that support expands to 55px, so a stub too short to have a direction of its own borrows one from the chain it has joined.

S1.3 Weights and scales of the pairing cost

The cost function itself is stated in the article, which also gives the reasoning behind its three deliberate properties. What follows are the values it is evaluated with.

Junction and gap links use separate weight sets, because a junction link is scored on direction alone while a gap link must also answer for the distance it spans. The curvature term carries a constant of 30, which places curvature, in px^{-1} , on the scale of the angular terms; without it a term in reciprocal pixels and a term in radians could not be added. Junction links set $w_\ell = 0$, a junction having no gap to pay for, and ℓ_0 and d_0 set the length scale of the gap penalty and the width of the chord fade respectively.

S1.4 Penalties, gates and admissibility

Leaving a stub unmatched carries a penalty rather than a prohibition, and the penalty p_i also fixes admissibility: an edge is a candidate only when $C_{ij} < p_i + p_j$. Exact maximum-weight partial matching then maximises $\sum(p_i + p_j - C_{ij})$ over disjoint pairs, which minimises total link cost plus p_i for every stub left free.

Table S1. Cost weights and scales of the released configuration. Junction and gap links are scored with separate sets; only the gap set carries a length term.

Symbol	Term	Junction	Gap
w_d	reversal angle θ	0.6	0.2
w_t	chord turns $\varphi_a + \varphi_b$	1.2	1
w_κ	curvature continuity	1	0.5
w_ℓ	length d/ℓ_0	—	0.15
ℓ_0	length scale (px)	—	60
d_0	chord-fade scale (px)	4	3
p_i	price of leaving a stub free	0.62	0.3

The admissibility limit is not a separate threshold. An edge with $C_{ij} \geq p_i + p_j$ contributes nothing positive to the objective, so a maximum-weight *partial* matching would never select it, and declining to offer it removes an edge that could not have been chosen.

At the configured penalties junction stubs carry $p_i = 0.62$, so a junction edge needs $C < 1.24$. Gap stubs carry $p_i = 0.30$, so a gap edge needs $C < 0.60$, and must in addition satisfy $d \leq 85$ px, $\theta \leq 0.40$ rad and $\varphi \leq 0.28$ rad at each end. The gap gates are the tighter set because a gap link asserts a continuation across evidence that is missing rather than merely ambiguous. The values were fixed on development scenes.

S1.5 The round schedule

Rounds $r = 0, \dots, 8 - 1$ apply a schedule fraction $\alpha_r = 0.85 + (1 - 0.85)r/7$ that scales the unmatched penalties and the two gap angle limits, and nothing else. Scaling a penalty *down* admits fewer edges, admissibility being $C_{ij} < p_i + p_j$, and scaling an angle limit down narrows the gap gates, so early rounds are the conservative ones and are deliberately under-linked: while the frames are least reliable, only pairings that survive the reduced penalties and narrowed gates are committed, and the constraints relax towards their configured values as the chains assembled so far improve the frame estimates. The final round, $\alpha_7 = 1$, is the only one run at the configured parameters, and is the one returned.

In full:

1. Skeletonise M , prune spurs, merge junction clusters into nodes, and store the remaining paths as arms with a stub at each end.
2. For $r = 0, \dots, 7$, re-solving both matchings from scratch and carrying only the chain structure forward from round $r - 1$:
 - (a) Re-fit every stub frame along the chain it now belongs to.
 - (b) At each node, form candidate edges between stubs of different arms with finite frames and cost, keep those admissible at α_r , and solve one maximum-weight matching per node.
 - (c) From the still-unmatched reliable stubs, form gap candidates on different arms and different nodes that pass the distance gate and the α_r -scaled angle gates, and solve one global maximum-weight matching over the admissible ones.
3. Return the links of round $r = 7$.

S1.6 Why the junction is solved as a whole

Joint and sequential solutions coincide at plain crossings and diverge only as nodes grow dense. Over 20 development scenes they never disagreed at nodes of degree 2–4 ($n = 542$), disagreed at 22 % of nodes with 5–8 stubs ($n = 187$) and at 69 % of nodes with 9 or more ($n = 233$), where the joint solution was better by a median of 0.90 in total cost.

That comparison holds one node’s costs fixed. Run inside the rounds, where each matching sets the frames the next is scored against, an early low-cost choice affects subsequent frame estimates and the difference is no longer confined to dense nodes.

S1.7 Instance output

Connected components of accepted arm links define instances. PLECTA paints every arm pixel and every junction cluster touched by a chain into that instance, so intersecting instances share crossing pixels. An isolated one-arm component shorter than 6 px is omitted. Accepted gap links determine identity but are not painted into the fixed centreline output. All effective parameters, including values previously inherited as code defaults, are materialised in the released JSON configuration.

S2 Depth stage: evidence and solver

This section gives the detail behind Section 2.6 of the article.

S2.1 Sampling geometry at a crossing

Where two instance centrelines intersect, a short arc of each is taken through the crossing (the shared *core*, a flat 7.5 px of arc, the same at every crossing), together with the *flanks* on either side of it. Flank samples are kept clear of the other strand’s stroke, because a sample taken inside it measures the union of two rods rather than one of them. A crossing with too few usable flank samples abstains rather than being decided on thin evidence.

The core length was previously sized from the two strands’ radii so that it spanned both silhouettes. It is now the constant above, and no per-rod radius enters the evidence path at all, so a radius estimate can no longer move a depth decision. Radii re-enter only downstream, in the non-interpenetration constraints that set metric height.

S2.2 One channel, and its noise floor

One quantity is read on core and flanks alike: median intensity. Write m_i and m_j for the two strands’ flank medians and m_c for the pooled core median. The evidence at the crossing is the score

$$s = 2 \frac{|m_c - m_j| - |m_c - m_i|}{\max(|m_i - m_j|, 2\hat{\sigma})}, \quad (\text{S1})$$

whose sign names the upper strand, the one whose flanks the core resembles more closely, and whose magnitude $|s|$ is the weight that decision carries. A crossing abstains when $|s| < 0.40$; the threshold is set directly in score units rather than as a band around a probability. Here $\hat{\sigma}$ is a noise estimate belonging to that crossing alone: 1.4826 times the median absolute deviation of the within-rod flank residuals, each strand’s flank samples being centred on their own median and the two residual sets pooled. It therefore measures the scatter within a rod’s own appearance, not the difference between the two rods.

A probability p is still written into the per-crossing record so that the output keeps its schema, but it is a monotone relabelling of s . It is neither a calibrated confidence nor the quantity anything is decided or weighted on, and should not be read as one.

Why the denominator is a noise floor

By the reverse triangle inequality the numerator of (S1) is bounded in magnitude by $|m_i - m_j|$, so a denominator equal to that separation drives the ratio to ± 1 whenever the two strands differ at all. The previous form of this rule floored the denominator at a small fixed constant in grey

levels, far below the noise of any of these images. The consequence was structural rather than statistical: a maximal score was then reachable on a separation smaller than the noise, and a maximal score exceeded any abstention threshold, so the rule could not abstain on such a crossing regardless of the evidence. Nor was that regime rare: a substantial fraction of crossings saturated this way, and among the saturated ones those whose separation fell below the noise were markedly less accurate than those separated by more than it. Flooring at $2\hat{\sigma}$ instead makes a confident score mean *separated by more than the noise* rather than merely separated, and restores the stage’s ability to abstain where it should.

The gradient channel, and why it was removed

An earlier version of this stage read a second quantity beside intensity: median gradient magnitude sampled on each strand’s own two edges at centreline $\pm r \mathbf{n}$, the axis gradient being near zero and therefore uninformative. The argument was that the upper strand’s edges stay crisp through the crossing while the lower strand’s are overwritten. It did not survive measurement. Standing alone the channel decided almost no real crossings, agreed with the intensity channel at chance, and was itself at chance; those readings were taken on the removed channel, and no configuration of the shipped code reproduces them, so they are recorded outside this manuscript rather than reported here. The reason is structural, and is visible in the channel’s own sampling geometry: within the core those $\pm r$ offsets fall inside the *other* strand’s stroke, so the quantity is contaminated exactly where it would have had to be read. The channel is removed, and with it the gradient image it required, the two channel weights that combined it with intensity, and the logistic, since with one channel there is nothing to combine and s is already the decision variable.

What the replacement is worth

The two rules were compared on 20 optically rendered synthetic scenes carrying 1342 crossings. The constants are a weaker claim: the flat core half-size (7.5 px) was settled by a sweep run on this same set, and the abstention threshold’s flatness was checked on it, so with respect to those two values the set is not independent. Both values are inherited rather than newly selected: 7.5 px is exactly what the previous rule’s radius-free fallback already computed, and both sit on a plateau rather than a peak, a flat core being no worse than oracle per-rod radii anywhere over 6–14 px and the accuracy unchanged across a band of thresholds around 0.40, so nothing below turns on either value, but the comparison is in-sample with respect to them.

At matched coverage, meaning both rules made to answer as often as the shipped one does at 83.0 %, the one-channel rule matches the two-channel rule’s accuracy: the difference is +0.0045, 95 % CI $[-0.0186, +0.0235]$, which is no difference. One reading goes against the replacement: forced to decide *every* crossing, including those it would decline, the one-channel rule is worse by -0.0142 , CI $[-0.0313, +0.0007]$. That is the expected direction, because abstention is exactly where an ordered confidence pays, and removing it removes the advantage; it is also not how either rule is run. What improves is the ordering of the stage’s own confidence. The area under the curve of correctness against $|s|$ rises from 0.6531 to 0.7586, a gain of +0.1056, CI $[+0.0659, +0.1425]$. That matters because $|s|$ is the edge weight the global ordering below is solved with: better-ordered confidence means contradictions are broken at the genuinely weakest relations rather than at arbitrary ones. It also converts into accuracy wherever more abstention is acceptable, by +0.0054 at 82.5 % coverage and +0.0121 at 80 %. The claim is therefore the same accuracy with better-ordered confidence, from a simpler evidence path.

S2.3 One order for the whole field

Each decided crossing contributes a directed edge weighted by $|s|$ itself, oriented toward the relation the sign of s names. The score is the primitive: the reported p is a relabelling of it and enters no computation here. Within each connected component of the crossing graph one total order over its instances is derived so as to maximise the total satisfied weight, and every relation in the component is then re-read from that order: the ones the order disagrees with are flipped rather than discarded, and both the flip and the original local score are recorded, to quantify how much the global correction moved. Which relations that turns out to be is the subject of the paragraph below.

Maximising satisfied weight is the minimum-weight feedback arc set objective, but only components of at most 14 instances attain it, by dynamic programming over subsets; larger ones get a greedy order refined with adjacent swaps, which carries no optimality guarantee. Because the order is re-derived for the component as a whole rather than only where relations conflict, a greedily ordered component can return a relation reversed that nothing contradicts, which an exact solution never does; a reversal lying on no cycle only loses weight. Fed ground-truth crossings with their true directions, input that is acyclic by construction and that an exact solver would leave untouched, the pass still reverses 24 relations across the 20 evaluation scenes, capping layer agreement at 0.886 where it should reach 1.000; a condense-then-order variant, which orders the condensation of the strongly connected components and reorders only inside genuine cycles, reverses none of them but measured worse on the actual noisy predictions and is not used in the released implementation.

S2.4 Layers and metric height

The corrected relations form a directed acyclic graph whose longest path fixes the smallest number of layers consistent with them; strands that never conflict share a layer, so the layer count follows from the ordering rather than from the number of instances. Each instance is then placed at a height respecting its order that keeps strands from interpenetrating at their measured radii, with the stack compacted so the film is as thin as those constraints allow.

S2.5 Configuration

Every tunable quantity of both stages (the cost weights and gates above, the core and window sizes, the abstention threshold in score units, and the exact-solver size limit) is carried in a single released parameter file, grouped by the stage it belongs to. Table S1 gives that file’s 2-D half; Table S2 gives its depth half. Every value in either table, and every parameter this section quotes, is taken from the released file.

S3 Measures

Every measure this paper reports is defined here, with what it rewards and its blind spot; none reads the rendered width of an instance.

The common-fragment set, shared by the first five measures below. The mask is skeletonised, spurs of at most 3 px are pruned, skeleton pixels with three or more 8-neighbours are deleted, the remaining 8-connected pieces are labelled, and pieces shorter than 6 px are discarded, so one fragment is one unambiguous arm between crossings. The partition reads neither the reference nor any prediction, so every method compared is cut into the same fragments. Each then takes a reference and a predicted identity by one rule, applied to both sides alike: the fragment is assigned to the instance whose mask overlaps the largest number of its pixels, the vote counted against each instance’s full mask rather than a flattened label image, so an instance that overlaps another keeps its pixels, provided that the winning overlap covers at least 30 % of

Table S2. The depth stage’s own entries in the released parameter file. The abstention threshold is in score units, not a band around a probability, and the core is flat: no per-rod radius enters the evidence path.

Quantity	Value	What it sets
core (px)	7.5	half-length of the shared crossing arc; flat at every crossing, so no per-rod radius enters the evidence path
window (px)	22	arclength sampled either side of a crossing for evidence
min. flank	6	fewer usable flank samples and the crossing abstains
abstain $ s $	0.4	abstention threshold, in score units, not a probability band
exact limit	14	components to this many nodes are ordered exactly, larger ones greedily

the fragment; failing that, it falls back to the most frequent nearest instance label within 6 px of the fragment, read from a deterministically flattened field; failing both, it stays unassigned and is scored as its own singleton cluster, on the reference side and the prediction side equally, so every fragment enters the scored partition. Ties at any step resolve to the lowest instance id, and the flattening order is fixed, so the assignment is deterministic. Because junction pixels are deleted first, no fragment contains one: none of these measures can see whether an emitted crossing pixel is given one owner or two, only whether the arms meeting there are grouped as the reference groups them.

Sensitivity to the scorer’s constants. Because the four constants above are shared with the method (the article’s metrics section), the stored predictions of all four methods were re-scored under one-at-a-time perturbations of each constant, with the grid frozen in writing before any result was computed: spur pruning 2–4 px, minimum fragment length 4–8 px, overlap threshold 0.20–0.50, fallback distance 3–9 px. The perturbations affect scoring only; every stored prediction and every configuration predates them. Table S3 reports PLECTA’s mean F_1 and its plain paired margins over each comparator at every perturbation, over the 50 degraded comparison scenes. Across the 10 combinations PLECTA’s mean F_1 spans 0.71–0.86, and the smallest paired margin over any comparator under any perturbation is 0.11. The level moves only with the minimum fragment length, and in the same direction for every method, because admitting shorter fragments adds hard, low-evidence units for everyone; the other three constants move the mean by less than 0.002, and no perturbation changes the ordering of the four methods.

Common-fragment pairwise F_1 , the primary endpoint. A pair of fragments is a true positive when it shares both identities, a false positive when the prediction joins fragments of different reference strands, a false negative when it separates fragments of one:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}. \quad (S2)$$

A clustering measure of the Rand family [1–3], permutation-invariant in the instance labels; it rewards holding a strand together across every crossing and gap it passes. Blind spots: it is quadratic in fragmentation, so a strand arriving in n pieces carries $\binom{n}{2}$ pairs and a broken mask is charged many times for one missed join, and it weights long, much-crossed strands far above short ones.

Adjusted Rand index [4] counts the same pairs corrected for chance, which matters because with a hundred strands per scene almost every pair genuinely belongs to different strands and the uncorrected index sits near 1 for anything. Blind spot: redundancy, since over the comparison set it never departs from pairwise F_1 by more than 0.02, so it confirms rather than adds.

Variation of information [5], in bits, lower better, is the conditional-entropy distance between the partitions: $VI_{\text{split}} = H(\text{prediction} \mid \text{reference})$ is paid for cutting one strand into

Table S3. Scorer-constant sensitivity. Stored predictions re-scored under one-at-a-time perturbations of the four common-fragment constants; Δ columns are plain paired mean margins, PLECTA minus comparator, over the scenes both methods completed. Sensitivity reporting only; the published constants were not changed in response.

Scorer constants	PLECTA F_1	Δ greedy	Δ DNai	Δ GraFT
published constants	0.82	0.15	0.48	0.34
spur prune 2 px	0.82	0.15	0.48	0.34
spur prune 4 px	0.82	0.15	0.48	0.34
min fragment 4 px	0.71	0.11	0.39	0.27
min fragment 8 px	0.86	0.15	0.51	0.36
overlap threshold 0.20	0.82	0.15	0.48	0.34
overlap threshold 0.40	0.82	0.15	0.48	0.34
overlap threshold 0.50	0.82	0.15	0.48	0.34
fallback distance 3 px	0.82	0.15	0.48	0.34
fallback distance 9 px	0.82	0.15	0.48	0.34

several, $VI_{\text{merge}} = H(\text{reference} \mid \text{prediction})$ for fusing several into one, and reporting both is what separates a method that divides from one that fuses. Blind spot: each component alone is degenerate, since a method merging everything never splits anything and scores a perfect zero on the split component, so neither is ever quoted singly.

Detection F_1 , the measure DNai reports, matches predicted to reference *objects* greedily by highest intersection over union, stopping below 0.1, DNai’s own shipped threshold, with precision over predicted and recall over reference objects. It uses a symmetric 2px placement tolerance, under which a reference pixel counts as found when the prediction passes within it and vice versa, the intersection being the smaller of the two counts. The tolerance is necessary: two independently read one-pixel centrelines two pixels apart intersect in *nothing*, so a strict overlap reports placement rather than shape, a problem the thin-structure benchmark FISBe states as a design rationale [6]. What keeps that tolerance a correction rather than an inflation is measurable, and the real fields supply the control: on their manual-derived axes, where the input mask is a skeleton of the annotation itself and nothing is displaced, the tolerance moves detection F_1 from 0.93 to 0.93 on B58_110, against 0.18 rising to 0.60 on the nnU-Net axis of the same field. It pays where two readings disagree in placement and nowhere else. Counting objects makes it the most forgiving reading of a fragmented mask and the least discriminative measure here. Blind spot: an arm swap, which it scores 1.000 where the pairwise score gives 0.000.

Crossing Fidelity states the contribution directly. At every junction node the reference induces a partition of the incident arms, two together exactly when the reference gives them the same strand, and so does the prediction; Crossing Fidelity is the fraction of crossings at which the two partitions are identical. Nodes are the junction clusters of the same pruned skeleton and a node’s arms are the fragments 8-adjacent to it. Only nodes whose arms carry at least two distinct reference strands are scored, since a node where every arm belongs to one strand is a skeletonisation artefact and scoring it would reward doing nothing. It is a study-specific diagnostic, not an established metric, though it follows measures that score correctness at branch points rather than per pixel [7, 8]. Blind spots, both of which govern its use: it is far less discriminative than the pairwise score, so it says what was solved and is never a ranking axis; and crossings carry no information about *gaps*, so it is silent on the cost the pairwise score pays for bridging them. Sharing its skeleton and assignment with the pairwise score makes it a sharper view of the same evidence rather than independent confirmation, and it is always printed beside its all-unpaired baseline.

Join F_1 counts decisions rather than pairs, for the real fields where the two mask sources fragment differently. A *decision* is a pair of distinct fragments whose endpoints come within 85px, the gap this method will bridge; reference and prediction each say join or not, and precision, recall and their harmonic mean are taken over decisions. Being linear rather than quadratic in fragmentation, and conditioned on the mask, it asks two mask sources comparably hard questions, and splitting and merging appear separately as low recall and low precision. Blind spot: a fragment pair the mask never brought within reach counts against nobody, so the measure is insensitive to a strand the mask lost entirely, and it is always quoted with the number of decisions each mask posed.

Shared-pixel precision and recall are precision and recall over the pixels the reference gives to more than one instance, computed on the raw instance masks. They exist because the pairwise measure discards junction pixels and is insensitive to whether a crossing pixel was marked as jointly owned at all: only a method emitting overlapping layers can score above zero on the recall, and one emitting a hard partition scores exactly zero by construction. Blind spot: neither component distinguishes a pixel shared for the right reason from one shared because a chain was painted with a whole junction cluster.

The runtime scaling exponent is the slope of an ordinary least-squares fit of log seconds on log foreground pixels, so an exponent b means cost grows as the b -th power of the amount of ink. It is reported instead of absolute time because these are independently optimised codebases, where absolute time measures implementation as much as algorithm. Blind spot: it describes a median trend and is silent on the tail, so it is quoted beside the per-density medians and the count of scenes stopped at the budget. Dropping those scenes biases the exponent downwards for any method whose slowest ones did not finish, which is why GraFT’s is a lower bound.

Two measures were computed and are not endpoints. GraFT’s *filament matched coverage* asks whether each reference strand was matched one-to-one to some detection, and PLECTA ranks *below* both comparators on it: at the densest stratum of GraFT’s own generator, 0.80 against 1.00 and 0.90. It is not an endpoint because it counts existence rather than correctness: a method emitting more objects than the scene contains has a spare for every strand, each method’s matched coverage equals the smaller of one and its emission ratio to within 0.16, and requiring a detection to cover half the axis it claims restores the order. *Centreline Dice* [9] was tried as an alternative matching criterion and not adopted: it tests each side’s skeleton against the other’s *volume*, and a centreline grouping has none, so it degenerates into the placement test the tolerance exists to correct. No cDice result is reported in this paper.

Admitting a reference at all. Every measure above compares a prediction with a reference, so a defective reference is invisible to all of them: it will simply score every method a little lower and change no ordering. A reference is therefore admitted only if it reproduces itself, that is, only if scoring it as though it were a prediction of itself returns a perfect score on every measure. Anything less means the reference disagrees with its own fragment assignment, and the shortfall bounds what any method could score against it.

S3.1 Statistical analysis

Synthetic-scene means and density strata are accompanied by 95% nonparametric bootstrap intervals (20,000 replicates; seed 20,260,810), resampled with stratification by density; the clean-minus-degraded analysis resamples paired scene differences within density. Per-coverage points in the comparison figures are plain means of ten scenes with no interval claimed. On the real SEM fields the inferential unit is the field, and the 2566 reference crossings entering the crossing measures are a count pooled over six condition-specific evaluations of the three fields, not a sample size. Active-component ablations are one-at-a-time development experiments on all 84 development scenes and do not reuse the evaluation set.

S4 Comparator configuration

The article’s comparator section states each adaptation and its cost; this section documents the mechanism behind each, so that the comparison can be verified. **DNAi** clusters junction pixels by single linkage and erases a disk at each cluster. At its released 10 px linkage distance, crossings closer than that chain into one cluster, which at 60 % coverage reaches 89 px across and is replaced by a single small erasure disk, leaving a whole tangle as one instance, which is why its shipped default is not a fair measure of it. Reducing the distance to 2 px on development scenes raised development F_1 and cut variation-of-information merge by about a factor of three, confirming the mechanism; the tuned configuration is the one reported. Two structural choices account for the residual difference: it commits to a pairing at every junction of degree two to four with no cost threshold, so it cannot decline a continuation the way a priced unmatched option can, and it attempts no junction above degree four, resolving 0.02 of the crossings above it. Those are about a tenth of the crossings, but they carry 37.2 % of same-strand fragment pairs on the degraded masks, one such junction severing every pair that spans it. **GraFT**’s errors are divisions rather than merges (variation-of-information split 1.17 against merge 0.39), which follows from a greedy longest-first path cover that cannot revise an early commitment once the scene collapses into one connected graph, as it does above 30 % coverage.

SIFNE enhances the image with an orientation filter transform, thresholds it by Otsu, thins it, and then works from the binary skeleton alone: it marks every pixel whose eight-connected neighbourhood sum reaches three as a crossing, erases each such pixel together with a square neighbourhood, and pairs the resulting fragment termini inside an angular search fan, bidirectionally and one pairing at a time [10]. Because everything after the thinning reads a binary axis, its own re-entry point is the thinned array, and the mask is injected there; every stage downstream is its released source unmodified. Only its graphical callbacks are re-expressed, so that it can run unattended, and its manual-correction stage is never invoked, since every comparator here runs without a human in the loop. One shipped default is degenerate at these densities and is re-selected on development scenes.

The erasure neighbourhood is 7 pixels square, applied at every pixel whose eight-connected neighbourhood sum reaches three, so the erasures multiply with the crossings: on scenes at 20, 40 and 60 % coverage the surviving mask is 51, 44 and 31 % of the input mask, and the evidence is removed before any pairing is attempted. On the sparse single-molecule fields the method was built for, the erased fraction is small. Selection was staged and both grids were fixed before either ran: first the three settings that govern how much mask survives, then the linkage gates at the winner of the first. Only the ten development scenes were read, never the comparison set. The chosen configuration reduces the erasure to 5 pixels square and the search radius to 35 pixels, raising development F_1 from 0.537 to 0.749 and crossing fidelity from 0.502, below the all-unpaired baseline, to 0.831 well above it. On the comparison set the same change is worth 0.55 to 0.71, and both are reported.

Two further points bear on reading its column. The search-fan angle turns out to be inert at the selected angular gate, because the gate is the stricter of the two tests for every fan wider than sixty degrees and therefore binds first; as verified in the source code. And the neighbourhood radius that its front end uses for orientation filtering is *not* inert once the mask is injected, because the same value sets the arclength over which every tip direction is estimated, which is the whole evidence its pairing sees.

Its crossing excision is not a scoring artefact. It deletes each crossing pixel and its neighbourhood and never restores them, but the common-fragment scorer also deletes junction pixels before labelling, and scores only whether the arms meeting there are grouped as the reference groups them. SIFNE links across the excised crossing, so it makes exactly the decision the endpoint measures. It does score zero on the shared-pixel measures, which is a structural property of a method that assigns no crossing pixel to anything.

Basu et al. is the one comparator with no released implementation: the only code its authors published covers the earlier centreline-localisation paper, at a departmental address that no longer resolves and that no archive captured. What is measured here is therefore a reimplementaion of the published principle rather than the authors' program, and it is reported as a reimplementaion baseline rather than as an external result. Only the geometric half of the method is reimplemented: a minimum spanning tree over the centreline pixels, deletion of every bifurcation node, and recombination of two- and three-segment groups scored by the curvature energy of a fitted smoothing cubic spline [11] less a bond energy, chosen by a set-partitioning integer program that is re-solved until the segment set stops changing. That half reads geometry alone and so accepts an axis mask directly. The other half, an intensity filter bank and a reverse diffusion that converges particles onto brightness crests, is not reimplemented, because scoring our own front end would measure the reimplementaion rather than the published method; this arm therefore evaluates the recombination logic only.

Its two user parameters, the maximum linking distance and the maximum tangent mismatch, were selected on the same ten development scenes and by the same criterion used for the others. Their published values, fixed in that paper for its own simulated benchmark, score 16 and 10 here only after selection; at the published pair the reimplementaion scores 0.076 on these masks, because a two-pixel linking distance admits almost no recombination on a mask whose gaps are wider than that. Both figures are recorded. Because the paper underdetermines two further quantities that change the result, the smoothing of the fitted spline and the quadrature of the bond energy, each was resolved in whichever direction favours the method, and the direction is recorded alongside the choice.

The comparison is read on crossing fidelity as well as on the pairwise score, and the two separate cleanly here: the recombination decides junctions almost as well as the exact matcher does, and loses the pairwise margin on the assembly built from those decisions, because a pairwise score over fragments is quadratic in fragmentation while a junction count is not.

S4.1 Margins the main text states without their intervals

The article's comparator section gives the crossing-resolution rates and the restricted DNAi comparison in summary. The paired figures are here.

Crossing Fidelity, paired against PLECTA over the degraded set: +0.065 [+0.047, +0.083] over the greedy baseline, +0.123 [+0.085, +0.165] over GraFT and +0.185 [+0.158, +0.211] over DNAi. On clean axes PLECTA resolves 0.83. Leaving every arm unpaired resolves 0.09 of these crossings, which is the all-unpaired baseline the rate is read against: a deterministic all-unpaired policy rather than a chance level.

Two things about that column belong with it. It is far less discriminative than the pairwise score, spanning 0.26 between best and worst method at 60 % coverage where pairwise F_1 spans 0.54, so it should be read as a count of what was solved and not as a ranking. Resolution rates decline for every method at the highest-degree nodes: pooled over the 3638 reference crossings of the degraded set, PLECTA partitions 0.81 exactly, but above degree four it resolves only 0.41, and no comparator is higher there.

The restricted DNAi comparison, which scores only the evidence a degree-limited, gap-blind method can reach, moves the clean-axis difference from +0.414 to +0.284 [+0.262, +0.307], still favouring PLECTA in 49 of 50 scenes.

S5 Depth measures

The measures of Section S3 score which centreline fragments belong to one instance. None of them can see depth: two reconstructions that agree on every instance but disagree about which

strand lies on top at every crossing are indistinguishable to all of them. The 2.5-D stage is therefore scored by its own family, defined here.

Crossing order accuracy. Each reference crossing names the instance physically on top; the reconstruction either names one or abstains. Three quantities are reported and none is quoted alone: *decided only*, the fraction of committed decisions that are correct, which by itself is not evidence because abstaining on every hard crossing scores 1.000; *all crossings*, which counts an abstention as an error and is the headline; and the *abstention rate* beside both. Chance is exactly 0.5, the decision being binary and symmetric, so only the margin above it carries information. A predicted crossing is credited only when it involves the same instance pair as a reference crossing and lies within the placement tolerance, so inventing crossings cannot help.

Depth order over instance pairs. Crossing accuracy is local. The global counterpart asks, for two instances, whether the reconstruction places the deeper one deeper, and is reported twice. Over *crossing pairs* it measures the stack where evidence exists. Over *all pairs* it includes instances that never cross and are therefore constrained by no evidence at all, whose relative depth is fixed by the solver’s tie-break rather than by the image. The two are never pooled, and the fraction of pairs sharing a component is reported beside them so the second can be read against what it could possibly measure. Both are invariant to monotone rescaling of depth, so they describe the arrangement rather than its units. Kendall’s τ over the same pairs is $2a - 1$ for accuracy a and is not reported separately.

Layer structure. Layers are an ordinal quotient of depth: a reconstruction can order every pair correctly yet cut the film into a different number of layers. Layer agreement, the fraction of instances placed in the reference layer, is therefore reported separately.

S6 Evaluation split and seeds

This section documents how every parameter was fixed on a separate development set and never on the scenes the article reports.

Development used 84 scenes spanning 20–60% coverage. Every parameter of the segmentation, every gate and price, and the round schedule were settled there, as was each comparator’s parameter translation, so that no method was tuned on the scenes it is scored on and all were tuned on the same ones.

The evaluation set comprises 128 scenes, 32 at each of 20, 30, 40 and 60% coverage.

The synthetic intervals describe sampling within this generator; the ranking is also tested on an independently published generator and on real SEM masks. The generator’s own shape parameters were also swept. Taking its bending stiffness from 50 to 150% of default, which moves the median centreline radius from 975 to 308 px, moved the level means of F_1 by at most 0.05 over 4 series, against a within-level scene-to-scene scatter of 0.05; taking the curve wavelength from 10 to 40 px moved them by at most 0.08, and the one series of 4 whose spread exceeded its own noise is non-monotonic with its minimum at the default, for which no mechanistic explanation was identified. Both sweeps are development-scene measurements, 10 scenes per level, reported as plain means.

S7 Manual annotation procedure

Reference annotations were created with an in-house interactive instance-segmentation tool for greyscale SEM images of crossing strands. Each bundle is stored as a separate mask, allowing

crossing pixels to belong to multiple strands. The saved annotation file is read directly by the evaluation without an intermediate export or conversion.

Tracing has two optional forms of assistance. Strokes can snap to the local intensity ridge, with polarity calibrated from the existing annotation, and can be re-rendered using a smoothing spline whose centreline shifts to the measured intensity crest only where that crest is consistent. Apparent widths are measured from perpendicular full-width-at-half-maximum profiles, referenced to the local minima on either side of the peak. The reported width is twice the nearer half-width, limiting inflation when a crossing strand merges into the profile. These widths do not enter the centreline-based segmentation scores.

The tool suggests gap bridges between collinear endpoints of different strands, but each suggestion requires the annotator’s acceptance. Strand identity through crossings and gaps therefore remains a manual decision.

References

- [1] William M. Rand. Objective criteria for the evaluation of clustering methods. *Journal of the American Statistical Association*, 66(336):846–850, 1971. doi:[10.1080/01621459.1971.10482356](https://doi.org/10.1080/01621459.1971.10482356).
- [2] Enrique Amigó, Julio Gonzalo, Javier Artiles, and Felisa Verdejo. A comparison of extrinsic clustering evaluation metrics based on formal constraints. *Information Retrieval*, 12(4):461–486, 2009. doi:[10.1007/s10791-008-9066-8](https://doi.org/10.1007/s10791-008-9066-8).
- [3] Ignacio Arganda-Carreras, Srinivas C. Turaga, Daniel R. Berger, Dan Cireşan, Alessandro Giusti, Luca M. Gambardella, Jürgen Schmidhuber, Dmitry Laptev, Sarvesh Dwivedi, Joachim M. Buhmann, Ting Liu, Mojtaba Seyedhosseini, Tolga Tasdizen, Lee Kamentsky, Radim Burget, Václav Uher, Xiao Tan, Changming Sun, Tuan D. Pham, Erhan Bas, Mustafa G. Uzunbas, Albert Cardona, Johannes Schindelin, and H. Sebastian Seung. Crowdsourcing the creation of image segmentation algorithms for connectomics. *Frontiers in Neuroanatomy*, 9:142, 2015. doi:[10.3389/fnana.2015.00142](https://doi.org/10.3389/fnana.2015.00142).
- [4] Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985. doi:[10.1007/BF01908075](https://doi.org/10.1007/BF01908075).
- [5] Marina Meilă. Comparing clusterings—an information based distance. *Journal of Multivariate Analysis*, 98(5):873–895, 2007. doi:[10.1016/j.jmva.2006.11.013](https://doi.org/10.1016/j.jmva.2006.11.013).
- [6] Lisa Mais, Peter Hirsch, Claire Managan, Ramya Kandarpa, Josef Lorenz Rumberger, Annika Reinke, Lena Maier-Hein, Gudrun Ihrke, and Dagmar Kainmueller. FISBe: A real-world benchmark dataset for instance segmentation of long-range thin filamentous structures. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 22249–22259, 2024. doi:[10.1109/CVPR52733.2024.02100](https://doi.org/10.1109/CVPR52733.2024.02100).
- [7] Todd A. Gillette, Kerry M. Brown, and Giorgio A. Ascoli. The DIADEM metric: Comparing multiple reconstructions of the same neuron. *Neuroinformatics*, 9(2-3):233–245, 2011. doi:[10.1007/s12021-011-9117-y](https://doi.org/10.1007/s12021-011-9117-y).
- [8] Elizabeth P. Reilly, Jeffrey S. Garretson, William R. Gray Roncal, Dean M. Kleissas, Brock A. Wester, Mark A. Chevillet, and Matthew J. Roos. Neural reconstruction integrity: A metric for assessing the connectivity accuracy of reconstructed neural networks. *Frontiers in Neuroinformatics*, 12:74, 2018. doi:[10.3389/fninf.2018.00074](https://doi.org/10.3389/fninf.2018.00074).
- [9] Suprosanna Shit, Johannes C. Paetzold, Anjany Sekuboyina, Ivan Ezhov, Alexander Unger, Andrey Zhylyka, Josien P. W. Pluim, Ulrich Bauer, and Bjoern H. Menze. cDice — a novel topology-preserving loss function for tubular structure segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16560–16569, 2021. doi:[10.1109/CVPR46437.2021.01629](https://doi.org/10.1109/CVPR46437.2021.01629).
- [10] Zhen Zhang, Yukako Nishimura, and Pakorn Kanchanawong. Extracting microtubule networks from superresolution single-molecule localization microscopy data. *Molecular Biology of the Cell*, 28(2):333–345, 2017. doi:[10.1091/mbc.E16-06-0421](https://doi.org/10.1091/mbc.E16-06-0421).
- [11] Paul Dierckx. *Curve and Surface Fitting with Splines*. Monographs on Numerical Analysis. Oxford University Press, Oxford, 1993. ISBN 978-0-19-853441-9.